

Arithmetic Structure of the Standard Model Mass Spectrum and the Bost-Connes Algebra

Shao-Wu Hu

Independent Researcher, Shanghai, China

forest@xindong.com ORCID: 0009-0009-8300-4691

Abstract

The 12 fermion masses of the Standard Model span 12 orders of magnitude, previously regarded as independent experimental inputs. Starting from the Bost-Connes algebra, this paper proves that the CRT decomposition of the character group $(\mathbb{Z}/7\mathbb{Z})^*$ uniquely determines the six fermion sectors and the three-generation structure, and provides a unified mass formula that reduces the 12 masses to the values of 4 Dirichlet L -functions on the critical line $s = 1/2$. All 9 charged fermion deviations are $\leq 2\%$. The three-generation absolute neutrino masses $m_1 = 5.1$, $m_2 = 10.0$, $m_3 = 50.6$ meV and the mass ratio $R = 34.5$ are new predictions of this paper, where m_1 and m_2 have not yet been directly measured.

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1 Introduction

The Standard Model is experimentally extremely successful, but its fermion mass spectrum—the three-generation lepton masses, the six quark masses, and the CKM and PMNS mixing parameters—are still input as free parameters rather than derived from within the theory [6, 22]. This paper proposes that these parameters are not mutually independent, but rather unfold layer by layer from a single arithmetic input—the prime

$$p = 7.$$

Specifically, the Bost–Connes (BC) algebra [1, 2, 12]

$$\mathcal{A} = C^*(\mathbb{Q}/\mathbb{Z}) \rtimes \mathbb{N}^*$$

at $p = 7$ has a local factor satisfying

$$(\mathbb{Z}/7\mathbb{Z})^* \cong \mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z},$$

where $\mathbb{Z}/2\mathbb{Z}$ encodes the doublet structure and $\mathbb{Z}/3\mathbb{Z}$ encodes the three-generation cycle. This paper will prove that the key hierarchy in the Standard Model mass spectrum can be systematically derived from this local structure.

The conclusions throughout are labeled by proof level: **[A]** (pure mathematical theorem or rigorous algebraic derivation), **[A*]** (numerically well-verified but with certain dynamical bridges still to be rigorously established), **[B]** (physical interpretation or model-based inference). A few physically motivated inferences are labeled **[B⁺]** (stronger than **[B]** but not yet fully rigorous).

2 Particle species: from six characters to six fermion sectors

At the prime $p = 7$, the multiplicative group

$$(\mathbb{Z}/7\mathbb{Z})^* = \{1, 2, 3, 4, 5, 6\}$$

is a cyclic group of order six. Taking 3 as a primitive root,

$$(\mathbb{Z}/7\mathbb{Z})^* \cong \mathbb{Z}/6\mathbb{Z}.$$

Furthermore, by the Chinese Remainder Theorem,

$$\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}.$$

This is the starting point of the entire paper: a minimal finite group that simultaneously carries a parity-type binary splitting and a generation-type ternary splitting.

The dual group $(\widehat{\mathbb{Z}/7\mathbb{Z}})^*$ also has six characters, denoted χ_n , $n = 0, 1, \dots, 5$, satisfying

$$\chi_n(3^k) = e^{2\pi i n k / 6}.$$

The orders of the characters are respectively

$$1, 6, 3, 2, 3, 6.$$

In this paper, the “order” is interpreted as a complexity level: the higher the order, the longer the multiplicative period required to return to the identity, and thus the more complex and strongly coupled the corresponding sector. This gives the following hierarchy:

Order of character	Complexity	Particle sector	Description
1	Simplest	Photon / trivial gauge sector	Trivial vacuum representation
2	Real binary	Neutrino	Self-conjugate, Majorana type
3	Three-cycle	Lepton	Phase-triple structure
6	Maximal complexity	Quark	Carries both $\mathbb{Z}/2$ and $\mathbb{Z}/3$

Here the “photon” is not one of the six fermion sectors, but rather labels the massless trivial gauge sector corresponding to the trivial character; the six genuine non-trivial fermionic degrees of freedom are given by the remaining character orbits and conjugate pairings. More precisely, the particle species are not “assigned” by hand, but are uniquely selected by the following four axioms.

Definition 2.1 (Four classification axioms). We consider organizing the six group elements or six characters into the six fermion sectors of the Standard Model, subject to:

- (1) **Conjugation closure**: each physical sector must be closed under the involution $a \mapsto 7 - a$;
- (2) **Three-generation compatibility**: the grouping of sectors must be compatible with $\mathbb{Z}/3\mathbb{Z}$ orbits;
- (3) **Complexity monotonicity**: the order of the character must correspond monotonically to physical complexity: order 2 is simpler than order 3, which is simpler than order 6;
- (4) **Real/complex phase separation**: the unique real non-trivial character (the Legendre symbol) must correspond separately to the self-conjugate sector and must not be mixed with the complex cubic characters.

Remark 2.2 (Why “higher order = more complex”? [A]). The statement in axiom (3) that “higher character order $\rightarrow$ more complex physics” is not an arbitrary assumption, but a natural consequence of the KMS states and L -function structure of the BC system:

- **Order 1** (χ_0 , trivial character): completely immune to KMS phase transitions; BC state is trivial at all temperatures $\rightarrow m = 0$ (gauge-protected!) $\rightarrow$ physically simplest;
- **Order 2** (χ_3 , Legendre character): real character, Gauss sum purely imaginary $\rightarrow$ tree-level $m = 0 \rightarrow$ second simplest;
- **Order 3** (χ_2, χ_4): complex character, Jacobi phase non-trivial, mass determined by phase structure $\rightarrow$ intermediate complexity;
- **Order 6** (χ_1, χ_5): contains all CRT directions, richest KMS state structure $\rightarrow$ most complex.

Therefore complexity monotonicity is not an externally imposed assumption, but a natural consequence of the KMS states and L -function Euler product structure of the BC algebra.

On the surface, assigning six objects to six categories has $6! = 720$ labeling possibilities; but these 720 “ambiguities” are completely eliminated by the four axioms above.

Theorem 2.3 (Uniqueness theorem, [A]). *Under the four axioms above, the assignment of the six characters of $(\mathbb{Z}/7\mathbb{Z})^*$ to the six Standard Model fermion sectors is unique up to renaming isomorphism. More specifically: the unique order-2 real character must correspond to the neutrino sector; the unique pair of order-3 conjugate characters must correspond to the lepton sector; and the two pairs of order-6 characters correspond to the two quark sectors. Therefore all 720 permutations collapse to a unique physical solution under the axiom constraints.*

Proof. First consider the order structure. Among the six characters, there is only one trivial character of order 1; only one non-trivial real character of order 2; one conjugate pair of order-3 characters; and two conjugate pairs of order-6 characters. Therefore by the complexity monotonicity of axiom (3), the three complexity levels of order 2, 3, and 6 are already uniquely fixed.

Next, axiom (4). The non-trivial real character is uniquely the Legendre character, which is self-conjugate and therefore can only correspond to a sector that is its own antiparticle; among fermions, this uniquely corresponds naturally to the Majorana-type neutrino sector. Thus the order-2 level uniquely falls to neutrinos.

For the order-3 level, there is only one conjugate pair, and its phase closes on cubic roots of unity, most naturally matching the three-generation periodicity of $\mathbb{Z}/3\mathbb{Z}$; if assigned to quarks, the quarks would lose the maximal complexity of simultaneously carrying $\mathbb{Z}/2$ and $\mathbb{Z}/3$, violating axiom (3). Hence order 3 uniquely corresponds to the lepton sector.

The remaining two pairs of order-6 characters automatically correspond to the quark sector. Since axiom (1) requires conjugation closure and axiom (2) requires compatibility with three-generation orbits, any attempt to exchange “lepton/quark” or “neutrino/lepton” would break one of: the order structure, self-conjugacy, or three-generation phase compatibility. Thus all allowed assignments are unique up to automorphisms and physical relabeling. $\square$

Therefore, the assignment of particle species is uniquely determined by the character structure of $p = 7$.

3 Three generations, spin and doublet structure

The CRT decomposition

$$\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$$

has the physical meaning that each state carries both a binary index and a ternary index. On $(\mathbb{Z}/7\mathbb{Z})^*$, the most natural binary operation is the involution

$$\iota(a) = 7 - a.$$

It satisfies $\iota^2 = \text{id}$, and thus generates a $\mathbb{Z}/2\mathbb{Z}$ action. Grouping the six elements into pairs under this involution yields three doublets:

$$D_1 = \{1, 6\}, \quad D_2 = \{2, 5\}, \quad D_3 = \{3, 4\}.$$

These three doublets correspond precisely to the three generations.

Proposition 3.1 (Doublets and three generations, $[\mathbf{A}]$). *The set $\{D_1, D_2, D_3\}$ is the complete set of orbits of $(\mathbb{Z}/7\mathbb{Z})^*$ under the involution $\iota(a) = 7 - a$; thus the three-generation structure is not an additional input, but the unique three-orbit decomposition given by the involutive arithmetic of $p = 7$.*

We now introduce the most central quantity in this paper: the doublet tension. For a doublet $D = \{a, 7 - a\}$, define

$$T(D) := |a - (7 - a)| = |2a - 7|.$$

Computing directly for the three doublets:

$$T(D_1) = |1 - 6| = 5, \quad T(D_2) = |2 - 5| = 3, \quad T(D_3) = |3 - 4| = 1.$$

The tension sequence is therefore

$$\{5, 3, 1\}.$$

Theorem 3.2 (Arithmetic progression of tensions, [A]). *For the three involution doublets of $p = 7$,*

$$D_1 = \{1, 6\}, \quad D_2 = \{2, 5\}, \quad D_3 = \{3, 4\},$$

the tension sequence satisfies

$$T(D_1), T(D_2), T(D_3) = 5, 3, 1,$$

forming an arithmetic progression with common difference 2.

Proof. For $D_k = \{k, 7 - k\}$, we have

$$T(D_k) = |k - (7 - k)| = |2k - 7|.$$

Substituting $k = 1, 2, 3$ yields 5, 3, 1. The difference between consecutive terms is 2, so they form an arithmetic progression with common difference 2. $\square$

The three doublets $D_1 = \{1, 6\}$, $D_2 = \{2, 5\}$, $D_3 = \{3, 4\}$ are given by $\mathbb{Z}/3\mathbb{Z}$ Galois orbits. Each doublet corresponds to one generation of charged leptons. The specific mass assignment ($D_1 \rightarrow e$, $D_2 \rightarrow \mu$, $D_3 \rightarrow \tau$) is determined by the Koide formula in Section 5.

3.1 Discrete spin and fermionic statistics

The doublet structure not only yields the three generations and stability hierarchy, but also naturally leads to the $\mathbb{Z}/2\mathbb{Z}$ classification of fermions/bosons and the arithmetic origin of spin.

Theorem 3.3 (Discrete spin, [A]). *Let $g = 3$ be a generator (of order 6) of $(\mathbb{Z}/7\mathbb{Z})^*$. By Euler's criterion,*

$$g^{(p-1)/2} = 3^3 = 27 \equiv -1 \pmod{7}.$$

That is, a “half-turn” $= g^3 = -1$. On the antisymmetric representation of the doublet $\{a, 7 - a\}$, -1 acts as $-\text{id}$:

$$g^3|\psi\rangle = -|\psi\rangle \quad (\text{fermionic state}), \quad g^6|\psi\rangle = +|\psi\rangle \quad (\text{full turn returns to original}).$$

In the discrete framework of this paper, this is the definition of spin 1/2: half-turn changes sign, full turn returns to the original.

Proof. $3^3 = 27 = 4 \times 7 - 1 \equiv -1 \pmod{7}$, which is a direct consequence of Euler's criterion. The involution $\iota: a \mapsto 7 - a$ on the doublet acts precisely as multiplication by -1 . The antisymmetric state satisfies $\iota|\psi\rangle = -|\psi\rangle$, i.e., half-turn changes sign; $\iota^2 = \text{id}$ means full turn returns to the original. $\square$

Theorem 3.4 (Fermion/boson $\mathbb{Z}/2\mathbb{Z}$ classification, [A]). *The eigenvalue decomposition of the involution ι on the doublet gives:*

- *Symmetric state* ($\iota = +1$): $a + (7 - a) = 7$ (constant, no information $\rightarrow$ bosonic statistics);
- *Antisymmetric state* ($\iota = -1$): $a - (7 - a) = 2a - 7 = \pm T$ (tension, has information $\rightarrow$ fermionic statistics).

The Legendre character χ_3 corresponds precisely: $\chi_3(\iota) = +1$ (QR) $\leftrightarrow$ boson; $\chi_3(\iota) = -1$ (QNR) $\leftrightarrow$ fermion.

Remark 3.5. Conventionally, spin $1/2$ is defined via continuous rotations of $SU(2)$ ($\pi_1(SO(3)) = \mathbb{Z}/2\mathbb{Z}$). In the framework of this paper, spin is given directly by the $\mathbb{Z}/2\mathbb{Z}$ subgroup $\{1, -1\}$ of $(\mathbb{Z}/7\mathbb{Z})^*$, without the need to introduce continuous groups. The two descriptions are equivalent in physical predictions.

Remark 3.6 (Half-integer spin and quadratic field extension). $-1 \equiv 6 \pmod{7}$ is not a quadratic residue ($\sqrt{-1}$ does not exist in $\mathbb{F}_7$) [15]. Half-integer spin requires the quadratic extension $\mathbb{F}_{7^2}$, corresponding to the double cover $SO(3) \rightarrow SU(2)$. This is the arithmetic origin of the spin-statistics theorem.

4 Jacobi phase, Cabibbo angle and CKM

The core parameters of the mass spectrum and CKM mixing are the Jacobi phase δ_2 and the Cabibbo angle λ . This section derives δ_2 from the Jacobi sum of the BC algebra, derives λ from δ_2 , and shows how λ gives the complete CKM matrix through the doublet jump-count expansion. Section 5 will use the same set of parameters to derive all 9 fermion masses.

Jacobi phase δ_2 [A]. The Jacobi sum $J(\chi_2, \chi_2) = \tau(\chi_2)^2 / \tau(\bar{\chi}_2)$ [15] measures the multiplicative autocorrelation of the character χ_2 with itself. Its phase $\arg(J)$ encodes the relative rotation angle between the three-generation doublets. The Eisenstein decomposition gives the exact value:

$$J(\chi_2, \chi_2) = -1 - d_2 \omega_3, \quad \omega_3 = e^{2\pi i/3},$$

where $d_2 = 3$. Therefore $\arg J = -\arctan(d_2^{3/2}) = -\arctan(3\sqrt{3}) = -79.11^\circ$, giving

$$\delta_2 = \frac{\arg J(\chi_2, \chi_2)}{\varphi(p)} = \frac{-\arctan(d_2^{3/2})}{2d_2} = -13.18^\circ. \quad (1)$$

QED correction: physical value $\delta = \delta_2 + \arcsin(\alpha_{\text{EM}}) = -12.74^\circ$. The correction $\Delta\delta = \arcsin(\alpha_{\text{EM}})$ arises from the first-order correction of the virtual process of the EM character χ_2 (self-loop $\chi_2^2 = \bar{\chi}_2$) to the Koide phase, whose magnitude is determined by the electromagnetic coupling constant α_{EM} [A*].

In the BC algebra, the multiplicative semigroup $\mathbb{N}^*$ enters the system through a Hecke-type action; on the local factor $(\mathbb{Z}/7\mathbb{Z})^*$, the most natural non-trivial action is given by multiplication by 2. Its action on the three doublets is

$$2 \cdot D_1 = D_2, \quad 2 \cdot D_2 = D_3, \quad 2 \cdot D_3 = D_1,$$

because

$$\begin{aligned} 2 \cdot \{1, 6\} &= \{2, 12\} \equiv \{2, 5\} = D_2, \\ 2 \cdot \{2, 5\} &= \{4, 10\} \equiv \{4, 3\} = D_3, \end{aligned}$$

$$2 \cdot \{3, 4\} = \{6, 8\} \equiv \{6, 1\} = D_1.$$

Therefore the Hecke action forms a three-cycle on the doublet space.

Theorem 4.1 (Hecke three-cycle, $[\mathbf{A}]$). *Let S_m^* denote the Hecke-type action on the BC local factor. For $m = 2$, S_2^* induces a three-cycle on the doublet set $\{D_1, D_2, D_3\}$:*

$$D_1 \rightarrow D_2 \rightarrow D_3 \rightarrow D_1.$$

In particular, its minimal positive order is 3.

Proof. The direct calculation above gives $S_2^*(D_1) = D_2$, $S_2^*(D_2) = D_3$, $S_2^*(D_3) = D_1$. Therefore $S_2^{*3} = \text{id}$. Since neither S_2^* nor S_2^{*2} is the identity, the minimal positive order is 3. $\square$

This three-cycle is the arithmetic prototype of inter-generation transitions. This paper interprets the CKM matrix as the amplitude matrix of Hecke transitions between doublets: diagonal elements represent probability amplitudes for remaining in the original doublet, and off-diagonal elements represent amplitudes for jumping across doublets. Thus quark mixing is no longer an arbitrary 3×3 unitary matrix, but is embedded in a discrete three-cycle path.

4.1 Single-step tunneling and the Cabibbo parameter

Let δ_2 be the phase determined by the Jacobi sum of the order-3 character, using the same rigid phase input as for the lepton sector; the fundamental amplitude for a single-step transition is taken as

$$\lambda = \frac{\sin(2\delta_2)}{2}.$$

When

$$\delta_2 \approx 13.18^\circ$$

one obtains

$$\lambda \approx 0.222.$$

This is precisely the empirical value of the Cabibbo parameter [19].

Proposition 4.2 (Single-step tunneling, $[\mathbf{A}^*]$). *If the minimal inter-doublet transition is induced by the phase difference $2\delta_2$, then its single-step tunneling amplitude satisfies*

$$\lambda = \frac{\sin(2\delta_2)}{2} = 0.222,$$

in agreement with the experimental Cabibbo parameter $\lambda_{\text{exp}} \approx 0.2245$ at the one-percent level.

The tag $[\mathbf{A}^*]$ is used here because the arithmetic origin of δ_2 is itself rigorous, yet interpreting that phase directly as the inter-doublet tunneling angle of quarks still constitutes a dynamical bridge.

Remark 4.3 (Why $\sin(2\delta_2)/2$? $[\mathbf{A}]$). The functional form $\lambda = \sin(2\delta_2)/2$ admits a precise character-theoretic interpretation. Since $\chi_5 = \chi_2 \times \chi_3$ and $\chi_1 = \bar{\chi}_2 \times \chi_3$, the factor χ_3 cancels in the CKM matrix elements; consequently CKM mixing originates entirely from the interference between χ_2 (argument $-\delta_2$) and $\bar{\chi}_2$ (argument $+\delta_2$):

- $\sin(\delta_2)$ = projection amplitude of the u-type sector onto the mixing direction;
- $\cos(\delta_2)$ = projection amplitude of the d-type sector onto the mixing direction;
- the product $\sin(\delta_2) \cos(\delta_2) = \frac{1}{2} \sin(2\delta_2)$ = the superposition of the two projections, i.e. the off-diagonal CKM amplitude.

This explains why the result is $\sin(2\delta_2)/2$ rather than $\sin(\delta_2)$ or $\sin(2\delta_2)$: it is the *product* projection of the two sector types onto the mixing basis, with the factor $1/2$ arising from a trigonometric identity rather than from a free parameter.

4.2 Wolfenstein hierarchy as a doublet-hop expansion

In the Wolfenstein parametrization [5, 20],

$$V_{us} \sim \lambda, \quad V_{cb} \sim A\lambda^2, \quad V_{ub} \sim \lambda^3(\rho - i\eta).$$

In the BC framework, the Wolfenstein hierarchy corresponds to path lengths in the Hecke three-cycle: V_{us} is the one-step transition between adjacent doublets; V_{cb} is a two-step process crossing two doublets, naturally suppressed to λ^2 ; and V_{ub} is a complete three-cycle closure, hence cubic. If one averages over the combinatorial degeneracy of the three-step closure, one obtains the approximation

$$|V_{ub}| \sim \frac{\lambda^3}{3}.$$

Remark 4.4 (Precise origin of the $1/3$ factor, [A]). A three-step closed loop starting from any doublet and returning to itself admits three paths:

$$D_1 \rightarrow D_2 \rightarrow D_3 \rightarrow D_1, \quad D_2 \rightarrow D_3 \rightarrow D_1 \rightarrow D_2, \quad D_3 \rightarrow D_1 \rightarrow D_2 \rightarrow D_3.$$

Each path contributes amplitude λ^3 but carries the phase ω^k of the Hecke three-cycle ($\omega = e^{2\pi i/3}$, $k = 0, 1, 2$). Coherent superposition gives

$$\lambda^3(1 + \omega + \omega^2) = 0.$$

Complete cancellation. However, the closure condition selects the trivial Fourier mode $k = 0$ (returning to the starting point = trivial representation), whose projection normalization is $1/|\mathbb{Z}/3\mathbb{Z}| = 1/3$:

$$|V_{ub}| = \lambda^3 \times \frac{1}{|\mathbb{Z}/3\mathbb{Z}|} = \frac{\lambda^3}{3}.$$

Hence $1/3$ is the normalization coefficient of the Fourier projector $P_0 = (1 + T_2 + T_2^2)/3$, not a probabilistic average.

Theorem 4.5 (Wolfenstein hop-count law, [A*]). *In the Hecke three-cycle model, the Wolfenstein hierarchy is interpreted as a doublet hop-count expansion:*

$$V_{us} \sim \lambda^1, \quad V_{cb} \sim A\lambda^2, \quad V_{ub} \sim \frac{\lambda^3}{3}.$$

The exponents are precisely the shortest Hecke path lengths.

Remark. A single nearest-neighbor hop yields the minimal suppression λ ; the shortest path for a two-step process has length 2, giving suppression λ^2 ; a three-step process corresponds to a complete closed loop with three paths, and after averaging one obtains $\lambda^3/3$. A rigorous unitary-matrix construction still requires additional dynamical input, but the hierarchy exponents themselves are uniquely determined by the discrete path lengths. $\square$

If $[D_F, T_a] = 0$, addition and multiplication separate and the direction is massless (as for the photon χ_0); if $[D_F, T_a] \neq 0$, they are inseparable and the mass is non-zero. For the real character χ_3 the entanglement cancels exactly at tree level and the diagonal entry vanishes—corresponding to neutrinos; for the complex characters χ_2, χ_1 the entanglement does not self-cancel—corresponding to massive leptons and quarks.

Local seed and global propagation. The commutator $\mu_j^{(a)}$ defined above is a local quantity on $\mathbb{Z}/p\mathbb{Z}$. Locally, all non-trivial characters have the same entanglement strength: the Gauss sum satisfies $|\tau(\chi_j)| = \sqrt{p}$ —if one looks only at $\mathbb{Z}/7\mathbb{Z}$, all fermion masses are equal and there is no hierarchy. The mass hierarchy arises from propagation from local to global: propagating the conflict seeds on $\mathbb{Z}/p\mathbb{Z}$ to the entire $\mathbb{Z}$, different characters receive different modulations from all primes $q = 2, 3, 5, \dots$, giving rise to Dirichlet L -functions [23]:

$$L\left(\frac{1}{2}, \chi_j\right) = \sum_{n=1}^{\infty} \frac{\chi_j(n)}{\sqrt{n}}.$$

$s = 1/2$ is the symmetry axis of the functional equation $\Lambda(s, \chi) = \Lambda(1 - s, \bar{\chi})$, and also corresponds to the BC algebraic KMS phase-transition point $\beta = 1$ ($s = \beta/2$). The values $|L(\frac{1}{2}, \chi_j)|$ are no longer equal for different j : $|L_2| = 0.318$, $L_3 = 1.147$, $|L_1| = 0.858$. This inequality of moduli is precisely the origin of the mass hierarchy—differences in fermion masses arise from the different efficiencies with which conflict seeds propagate through the primes. Thus all fermion masses are given by the values of $L(\frac{1}{2}, \chi)$ on the critical line.

Standard Model fermion masses span 12 orders of magnitude—from the top quark at 173 GeV to neutrinos at ~ 0.05 eV. In the BC framework all fermion masses arise from different directions of the same character group $(\mathbb{Z}/7\mathbb{Z})^*$, so cross-species ratios are precisely determined by the ratios of L -functions at $s = 1/2$ for different characters:

- Lepton masses are completely determined by the modulus (absolute scale) and the phase (three-generation structure) of $L(\frac{1}{2}, \chi_2)$;
- d-type quarks start from leptons, multiplied by the color factor d_2 and the weak-sector weight $L(\frac{1}{2}, \chi_3)/\zeta(\frac{1}{2})$;
- u-type quarks involve all three non-trivial L -functions L_1, L_2, L_3 , reflecting the fact that $\chi_1 = \bar{\chi}_2 \cdot \chi_3$ contains all characters;
- neutrinos correspond to the unique real character χ_3 —tree-level mass is zero, and the non-zero mass arises from the scattering echo of χ_3 on the χ_0 background (§5.7).

Charged fermions (complex characters) acquire mass through a first-order Dirac process; neutrino (real character) mass arises from the dilution of the χ_3 signal across N^2 channels of the χ_0 background (§5.7). The 12 orders of magnitude in the mass hierarchy are a direct consequence of the real/complex nature of the characters.

The full formulae involve four L -function values, each with a clear physical role:

Symbol	Value	Physical role
$L_2 = L(\frac{1}{2}, \chi_2) $	0.318	Lepton-sector scale (modulus $\rightarrow m_\tau$, phase $\rightarrow$ Koide $\rightarrow$ three generations)
$L_3 = L(\frac{1}{2}, \chi_3)$	1.147	Weak-sector weight (d-type cross-species, u-type transition probability)
$L_1 = L(\frac{1}{2}, \chi_1) $	0.858	u-type sector scale (transition probability)
$\zeta = \zeta(\frac{1}{2}) $	1.460	Global normalization (trivial character)

Here $L_{01} \equiv L_2$, $L_{10} \equiv L_3$, $L_{11} \equiv L_1$ follow the CRT double-index notation (subscript ij denotes the $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ components).

5.1 Character paths

The local seed propagates from χ_0 (vacuum) through character space to the sector of each fermion. Which characters the path passes through determines which L -function values appear in the mass formula; the power of each L -factor is jointly determined by the Galois structure of that character and the coherence of the propagation.

Power rule. The power of each L -factor equals the Galois degree times the coherence degree:

Galois $\times$ Coherence	Coherent ($C=1$)	Independent ($C=2$)
Complex character ($G=2$)	2	4
Real character ($G=1$)	1	2

Galois degree: complex characters have a Galois pair ($\chi_k \leftrightarrow \bar{\chi}_k$), and taking the modulus squared gives $G = 2$; real characters are self-conjugate, so $G = 1$. Coherence degree: propagation within the same sector (e.g. χ_2 in the lepton sector) is coherent, $C = 1$; cross-sector propagation (e.g. χ_3 from leptons to d-type quarks) is independent Born, $C = 2$. Thus the power of $|L_2|$ in the lepton sector is $2 \times 2 = 4$ (complex $\times$ independent), and the power of L_3 in the b -quark is $1 \times 1 = 1$ (real $\times$ coherent), while in s and d quarks it is $1 \times 2 = 2$ (real $\times$ independent).

Leptons: $\chi_0 \rightarrow \chi_2$. Leptons couple only to the even character χ_2 (order 3, $\mathbb{Z}/3\mathbb{Z}$ direction). One-step propagation; the mass formula contains only L_2 :

$$m_\ell \propto |L_2|^4.$$

The power $4 = G \times C = 2 \times 2$: complex character ($G = 2$), unique non-trivial direction within the lepton sector ($C = 2$, independent Born). This is the shortest path and the simplest mass formula.

d-type quarks: $\chi_0 \rightarrow \chi_2 \rightarrow \chi_3$. d-type quarks (b, s, d) pass additionally through χ_3 (order 2, $\mathbb{Z}/2\mathbb{Z}$ direction, weak sector) on top of the lepton path. Two-step propagation; the mass formula contains both L_2 and L_3 :

$$m_d \propto |L_2|^4 \cdot L_3^{\text{power}}.$$

The power of L_3 varies by generation: for gen3 (b) the χ_3 crossing is coherent (amplitude level), giving L_3^1 ; for gen2/gen1 (s, d) the χ_3 is independent Born, giving L_3^2 . The mass hierarchy of d-type quarks is therefore modulated by an additional L_3 factor compared to leptons.

u-type quarks: $\chi_0 \rightarrow \chi_2 \rightarrow \chi_3 \rightarrow \chi_1$. The character $\chi_1 = \bar{\chi}_2 \cdot \chi_3$ of u-type quarks (c, t, u) contains information from all non-trivial characters. Three-step propagation; the mass formula contains all three L -values:

$$m_{u\text{-type}} \propto |L_2|^4 \cdot L_3^2 \cdot |L_1|^4.$$

The longest path, the most complex formula. $|L_1|^4$ likewise arises from the Born probability of a complex character ($G \times C = 2 \times 2$). u-type quarks “see” the entire $\mathbb{Z}/6\mathbb{Z}$

structure; this is the arithmetic reason why their inter-generation structure involves force coupling constants (rather than the pure Koide formula). The L -factors are the same for all three generations; all mass differences among the three generations arise from the inter-generation factors, with the Dyson self-loop for the top and the weak coupling for the u quark to be elaborated in the respective subsections.

Neutrinos: $\chi_0 \rightarrow \chi_2 \rightarrow [\chi_0 \text{ round-trip}] \rightarrow \chi_2$. The character χ_3 of neutrinos is real ($\chi_3 = \bar{\chi}_3$, order 2). The diagonal commutator of a real character vanishes at tree level: the direct channel is closed. A non-zero mass must be realized via an indirect path.

The character multiplication $\chi_3^2 = \chi_0$ provides the unique indirect channel: the χ_3 signal is mapped to the trivial character χ_0 via squaring, then returns via χ_0 back to χ_3 . Physically this corresponds to the neutrino field completing a Majorana-type round trip through the χ_0 (vacuum/KMS thermal bath) channel—the signal must “borrow the vacuum” to propagate because the direct channel has no propagation mode. χ_0 is the trivial character and carries $N = \varphi(p^4) = 2058$ KMS active modes; the signal is dispersed into N modes upon entry and dispersed again upon return.

The L -factors are doubled by the round trip:

$$m_\nu \propto |L_2|^8.$$

The lepton path contributes $|L_2|^4$; the χ_0 round trip multiplies it by another $|L_2|^4$, giving $|L_2|^8$.

The path length determines the complexity of the mass formula: the more character nodes the path passes through, the more varieties of L -factors and node normalization factors appear. Leptons pass through only χ_2 (one step); u-type quarks pass through all three non-trivial characters (three steps). The L -factors of neutrinos share the same origin as those of leptons (both are powers of $|L_2|$), but the N^2 dilution factor introduced by the χ_0 round trip causes 12 orders of magnitude of mass suppression.

5.2 Node factors

The path determines the L -factors in the mass formula. The mass of each fermion also requires the normalization weights of each character node along the path—the node factors. Node factors are determined by the algebraic structure of the CRT decomposition $\mathbb{Z}/6\mathbb{Z} = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$; physically they correspond to a single principle: *indistinguishable (symmetric) states are normalized at the amplitude level by $\div\sqrt{n}$; distinguishable (independent) states are summed at the probability level by $\times n$* .

Upon reaching character χ_k , three independent effects may be triggered:

1. **Galois projection:** complex characters (χ_2, χ_1) have Galois pairs ($\chi_k \leftrightarrow \bar{\chi}_k$) that are physically indistinguishable—symmetric states, amplitude-level normalization $\div\sqrt{d_1}$, $d_1 = [\mathbb{Q}(\zeta_7) : \mathbb{Q}(\zeta_7)^+] = 2$. Real characters (χ_3) are self-conjugate; no projection, factor $\times 1$.
2. **Color liberation:** χ_3 corresponds to the $\mathbb{Z}/2\mathbb{Z}$ direction in CRT; the three color states are independent in this direction—distinguishable states, probability-level $\times d_2 = \times 3$. Other characters do not trigger this.
3. **Background normalization:** when the path crosses from χ_2 into χ_3 , the contribution $\zeta(\frac{1}{2})$ of the trivial character χ_0 provides the baseline, and the crossing costs $\div|\zeta(\frac{1}{2})|$. Passing through χ_0 itself causes dilution by $N = \varphi(p^4) = 2058$ KMS active modes, each passage costing $\div N$.

χ_k	Galois projection	Color liberation	Background norm.	Node factor
χ_2	$\div\sqrt{d_1}$	$\times 1$	—	$1/\sqrt{2}$
χ_3	$\times 1$	$\times d_2$	$\div \zeta $	$3/1.460$
χ_1	$\times 1$	$\times 1$	—	1
χ_0	—	—	$\div N$	$1/2058$

Node coefficient application table. The node coefficient of each fermion equals the product of single-pass factors at each node along the path, raised to the number of passes:

Particle	χ_2 ($\div\sqrt{d_1}$)	χ_3 ($\times d_2/ \zeta $)	χ_1 ($\times 1$)	χ_0 ($\div N$)	Node coefficient
τ, μ, e	$\times 1$	—	—	—	$1/\sqrt{d_1}$
b	$\times 1$	$\times 1$	—	—	$d_2/(\sqrt{d_1} \zeta)$
s, d	$\times 1$	$\times 1$ (Born)	—	—	$1/ \zeta ^2$
c, t, u	$\times 1$	$\times 1$	$\times 1$	—	$1/\sqrt{d_1}$
ν	$\times 2$	—	—	$\times 2$	$d_1^{-1}N^{-2}$

Neutrinos pass through χ_2 twice ($\div\sqrt{d_1} \times 2 = \div d_1$) and χ_0 twice ($\div N \times 2 = \div N^2$). For s/d the χ_3 is in independent mode (Born, $C=2$), with the color liberation factor d_2 consumed on the second pass.

5.3 Unified mass formula

Combining the path and node factors, the mass of each fermion is assembled from three contributions:

$$m_f = v \times \underbrace{\prod_{\text{path}} |L_k|^{\text{power}}}_{\text{scale}} \times \underbrace{\prod_{\text{nodes}} \text{normalization}}_{\text{node factors}} \times \underbrace{\text{Gen}_f}_{\text{inter-gen. factor}}. \quad (3)$$

$v = 246.22 \text{ GeV}$ is the experimental Higgs VEV.

Physical meaning of the three factors.

- **Scale** (L -factors): the Born probability of the local conflict seed after global propagation on $\mathbb{Z}$. The path determines which characters are traversed, and the power rule determines the Born level of each step. All differences in mass scale between sectors arise from the unequal moduli of $|L_2|$, L_3 , and $|L_1|$ at $s = 1/2$.
- **Node factors**: the normalization weight assigned by the CRT structure of character space to each node. Galois symmetry ($\div\sqrt{d_1}$), color degeneracy ($\times d_2$), background baseline ($\div|\zeta|$), and KMS dilution ($\div N$) are all uniquely determined by the algebraic structure with no free parameters.
- **Inter-generation factor** (Gen): the mass splitting among the three generations within the same sector. For leptons and heavy quarks it is given by the Koide rotation K_k (§4), derived from the Jacobi phase δ_2 ; the lightest quarks use the Cabibbo transition probability λ^2 . The Dyson enhancement and weak coupling for u-type quarks, and the Hurwitz weights for neutrinos, are elaborated in the respective subsections.

Among the three factors, the scale and node factors are uniquely determined by the path (§5.1) and the CRT structure (§5.2); the inter-generation factor is the only distinguishing element among the three generations within the same sector. The unified formula is now expanded separately for the four fermion classes, giving the specific mass of each generation.

5.4 Lepton masses

The lepton path is $\chi_0 \rightarrow \chi_2$ (§5.1). The unified formula expands as:

$$m_\ell = v \times \underbrace{|L_2|^4}_{L\text{-factor}} \times \underbrace{\frac{1}{\sqrt{d_1}}}_{\text{node}} \times \underbrace{K_k}_{\text{inter-gen.}}$$

L-factor: $|L_2|^4$. Leptons couple only to the even character χ_2 ; the mass scale is determined by the modulus of $L(\frac{1}{2}, \chi_2)$. $|L_2|^4$ is the Born probability of the complex character ($G \times C = 2 \times 2$, the unique solution satisfying C^* -algebra positivity and normalization [24]). The Dedekind decomposition places it within the cyclotomic field $K = \mathbb{Q}(\zeta_7)$:

$$y_\tau = |L(\frac{1}{2}, \chi_2)|^4 = \left(\frac{\zeta_{K+(\frac{1}{2})}}{\zeta(\frac{1}{2})} \right)^2,$$

the contribution of the totally real subfield K^+ on the critical line.

Node factor: $1/\sqrt{d_1}$. The Galois pair ($\chi_2 \leftrightarrow \bar{\chi}_2$) of χ_2 is physically indistinguishable; symmetric states are normalized at amplitude level by $\div \sqrt{d_1}$, where $d_1 = [\mathbb{Q}(\zeta_7) : \mathbb{Q}(\zeta_7)^+] = 2$.

Inter-generation factor: Koide K_k . The three-generation mass ratios are given by the *phase* δ of $L(\frac{1}{2}, \chi_2)$ through the Koide rotation [3, 18]:

$$f_k = 1 + \sqrt{d_1} \cos(\delta + 120^\circ k), \quad K_k = (f_k/f_0)^2.$$

$\delta = -12.74^\circ$ (Jacobi phase δ_2 plus QED correction $\arcsin \alpha_{\text{EM}}$). $r = \sqrt{d_1} = \sqrt{2}$ yields the Koide formula $Q = 2/3$, valid for any CM field $\mathbb{Q}(\zeta_p)$.

Three-generation mass formulae.

$$m_\tau = v \times |L_2|^4 \times \frac{1}{\sqrt{d_1}} \times K_0 = \frac{|L_2|^4}{\sqrt{2}} \times v, \quad (4)$$

$$m_\mu = v \times |L_2|^4 \times \frac{1}{\sqrt{d_1}} \times K_1 = m_\tau \times (f_1/f_0)^2, \quad (5)$$

$$m_e = v \times |L_2|^4 \times \frac{1}{\sqrt{d_1}} \times K_2 = m_\tau \times (f_2/f_0)^2. \quad (6)$$

$|L_2| = 0.318$ (scale) and $\delta = -12.74^\circ$ (phase) control the absolute mass and the inter-generation ratios, respectively.

Particle	L -factor	Node	Gen	Prediction	Experiment	Deviation
τ	$ L_2 ^4$	$1/\sqrt{d_1}$	K_0	1791 MeV	1777 MeV	+0.8%
μ	$ L_2 ^4$	$1/\sqrt{d_1}$	K_1	106.6 MeV	105.7 MeV	+0.9%
e	$ L_2 ^4$	$1/\sqrt{d_1}$	K_2	0.512 MeV	0.511 MeV	+0.1%

Deviations for all three generations are $< 1\%$. The extreme hierarchy $m_\tau/m_e \approx 3500$ arises from the near-zero crossing $f_2 \rightarrow 0$.

5.5 d-type quark masses

The d-type quark path is $\chi_0 \rightarrow \chi_2 \rightarrow \chi_3$ (§5.1), corresponding to the Georgi–Jarlskog relation [4]. The unified formula expands as:

$$m_d = v \times \underbrace{|L_2|^4 \cdot L_3^{\text{pow}}}_{L\text{-factor}} \times \underbrace{\text{node}}_{\text{see below}} \times \underbrace{\text{Gen}}_{\text{inter-gen.}}$$

L -factor: $|L_2|^4 \cdot L_3^{\text{pow}}$. On top of the lepton factor $|L_2|^4$, d-type quarks additionally traverse χ_3 (weak sector). $L_3 = L(\frac{1}{2}, \chi_3) = 1.147$ is a real character and self-conjugate ($\chi_3 = \bar{\chi}_3$). The power of L_3 varies by generation: the b quark (gen3) shares the χ_2 direction with leptons, so the cross-sector traversal is at amplitude level, giving L_3^1 ; for s and d quarks χ_3 is independent Born, giving L_3^2 .

Node factors. b quark: the path passes through χ_2 (Galois $\div \sqrt{d_1}$) and χ_3 (color liberation $\times d_2$, background $\div |\zeta|$), giving node $= d_2/(\sqrt{d_1} |\zeta|) = 3/(\sqrt{2} \times 1.460)$. s/d quarks: the color liberation factor d_2 of χ_3 is consumed on the second pass (Born level), giving node $= 1/|\zeta|^2$.

Inter-generation factor. For b (gen3) Koide K_0 is used; for s (gen2) Koide K_1 ; for d (gen1) Cabibbo λ^2 . The inter-generation suppression of the d quark comes from inter-doublet hopping rather than the Koide rotation.

Three-generation mass formulae.

$$m_b = v \times |L_2|^4 L_3 \times \frac{d_2}{\sqrt{d_1} |\zeta|} \times K_0, \quad (7)$$

$$m_s = v \times |L_2|^4 L_3^2 \times \frac{1}{|\zeta|^2} \times K_1, \quad (8)$$

$$m_d = v \times |L_2|^4 L_3^2 \times \frac{1}{|\zeta|^2} \times \lambda^2. \quad (9)$$

Particle	L -factor	Node	Gen	Prediction	Experiment	Deviation
b	$ L_2 ^4 L_3$	$d_2/(\sqrt{d_1} \zeta)$	K_0	4219 MeV	4180 MeV	+0.9%
s	$ L_2 ^4 L_3^2$	$1/ \zeta ^2$	K_1	92.9 MeV	93.4 MeV	−0.5%
d	$ L_2 ^4 L_3^2$	$1/ \zeta ^2$	λ^2	4.58 MeV	4.67 MeV	−2.0%

Deviations for all three generations are $\leq 2\%$. The ratio $m_b/m_\tau = d_2 L_3/|\zeta| = 2.356$ (experiment 2.352, deviation +0.1%) equals the color number times the L -function ratio.

5.6 u-type quark masses

The u-type quark path is $\chi_0 \rightarrow \chi_2 \rightarrow \chi_3 \rightarrow \chi_1$ (§5.1). The unified formula expands as:

$$m_u = v \times \underbrace{|L_2|^4 \cdot L_3^2 \cdot |L_1|^4}_{L\text{-factor}} \times \underbrace{\frac{1}{\sqrt{d_1}}}_{\text{node}} \times \underbrace{\text{Gen}}_{\text{inter-gen.}}$$

L-factor: $|L_2|^4 L_3^2 |L_1|^4$. The longest path, containing all three L -values. Both $|L_2|^4$ and $|L_1|^4$ are Born probabilities for complex characters ($G \times C = 2 \times 2$); L_3^2 is the independent Born factor for a real character ($G \times C = 1 \times 2$). The character $\chi_1 = \bar{\chi}_2 \cdot \chi_3$ of u-type quarks contains all non-trivial characters, so u-type quarks “see” the entire $\mathbb{Z}/6\mathbb{Z}$ structure. Unlike the amplitude-level L_3^1 in d-type quarks, the u-type and lepton characters differ ($\chi_1 \neq \chi_2$), so the cross-sector factor reduces to Born probability level.

Node factor: $1/\sqrt{d_1}$. The path passes through χ_2 (Galois $\div \sqrt{d_1}$), χ_3 , and χ_1 . The χ_3 and χ_1 nodes are transparent (§5.2), giving a total node factor of $1/\sqrt{d_1} = 1/\sqrt{2}$, the same as for leptons.

Inter-generation factor. The inter-generation structure of u-type quarks involves force coupling constants, and differs among the three generations:

- c (gen2, reference): Koide K_0 , same type as the lepton τ ;
- t (gen3): Dyson enhancement $1/\alpha_{\text{EM}} \approx 137$ —the self-loop of the EM character χ_2 ($\chi_2^2 = \bar{\chi}_2$) allows infinite virtual scattering; summing the Born series gives the enhancement factor;
- u (gen1): Cabibbo λ^2 multiplied by the weak coupling constant $\alpha_W \approx 1/29.6$ —each inter-generation hop passes through the χ_3 direction, paying an additional weak probability compared to d-type.

Three-generation mass formulae.

$$m_c = v \times |L_2|^4 L_3^2 |L_1|^4 \times \frac{1}{\sqrt{d_1}} \times K_0, \quad (10)$$

$$m_t = v \times |L_2|^4 L_3^2 |L_1|^4 \times \frac{1}{\sqrt{d_1}} \times \frac{1}{\alpha_{\text{EM}}}, \quad (11)$$

$$m_u = v \times |L_2|^4 L_3^2 |L_1|^4 \times \frac{1}{\sqrt{d_1}} \times \lambda^2 \alpha_W. \quad (12)$$

The L -factors and node factor are the same for all three generations; all differences reside in the inter-generation factors.

Particle	L -factor	Node	Gen	Prediction	Experiment	Deviation
c	$ L_2 ^4 L_3^2 L_1 ^4$	$1/\sqrt{d_1}$	K_0	1273 MeV	1270 MeV	+0.2%
t	$ L_2 ^4 L_3^2 L_1 ^4$	$1/\sqrt{d_1}$	$1/\alpha_{\text{EM}}$	174453 MeV	172760 MeV	+1.0%
u	$ L_2 ^4 L_3^2 L_1 ^4$	$1/\sqrt{d_1}$	$\lambda^2 \alpha_W$	2.1 MeV	2.16 MeV	−1.9%

Deviations for all three generations are $\leq 2\%$. $m_t/m_c = 1/\alpha_{\text{EM}} \approx 137$: the anomalously large top mass is a direct consequence of the EM self-loop. $m_u/m_c = \lambda^2 \alpha_W \approx 1/600$: the extreme lightness of the u quark results from Cabibbo suppression multiplied by weak-force suppression.

5.7 Neutrino masses

The neutrino path is $\chi_0 \rightarrow \chi_2 \rightarrow [\chi_0 \text{ round-trip}] \rightarrow \chi_2$ (§5.1). The unified formula expands as:

$$m_\nu = v \times \underbrace{|L_2|^8}_{L\text{-factor}} \times \underbrace{\frac{1}{d_1 N^2}}_{\text{node}} \times \underbrace{\lambda_k}_{\text{inter-gen.}}$$

L-factor: $|L_2|^8$. The χ_0 round trip doubles the $|L_2|^4$ of the lepton path. Physically, the character χ_3 of neutrinos is real ($\chi_3 = \bar{\chi}_3$) and the direct channel is closed. $\chi_3^2 = \chi_0$ provides the unique indirect channel: the signal is mapped to the trivial character χ_0 via squaring and then returns via χ_0 back to χ_3 , equivalent to traversing the lepton path once more, giving $|L_2|^4 \times |L_2|^4 = |L_2|^8$.

Node factor: $1/(d_1 N^2)$. χ_2 is passed twice: Galois projection $(\div \sqrt{d_1})^2 = \div d_1$. χ_0 is passed twice: KMS dilution $(\div N)^2 = \div N^2$, $N = \varphi(p^4) = 2058$. Total node factor $= 1/(d_1 N^2)$. This is the primary reason for the extreme smallness of neutrino masses: dilution by $N^2 \approx 4.2 \times 10^6$ channels.

Inter-generation factor: Hurwitz weights λ_k . The neutrino inter-generation factor is not given by the Koide K_k but is determined by the eigenvalues of the Koide–Hurwitz mass matrix (unlike the traditional seesaw mechanism [10, 11, 21], no high energy scale M_R is introduced here).

The doublet product $V_k = a_k(p - a_k)$ satisfies the Fermat condition $V_k^3 \equiv -1 \pmod{7}$ [**A**]— $-1 = \text{involution} = \text{arithmetic expression of the Majorana condition}$, with $n = 3 = (p - 1)/2$ the minimal unifying exponent.

The Hurwitz ζ -function lifts the local Fermat structure to the global level:

$$w_k = \frac{1}{p^6} \zeta_H\left(3, \frac{a_k}{p}\right) \times \zeta_H\left(3, \frac{p - a_k}{p}\right).$$

The Koide circulant matrix $C = \text{circ}(f_0, f_2, f_1)$ combined with the Hurwitz weights gives the mass matrix:

$$\mathcal{M} = C \cdot \text{diag}(w_1, w_2, w_3) \cdot C^T.$$

C is shared with charged leptons (same Koide parameter $r = \sqrt{2}$ and phase δ), but $w_1 \neq w_2 \neq w_3$ breaks the $\mathbb{Z}/3\mathbb{Z}$ cyclic symmetry of Koide so that the eigenvalues λ_k deviate from simple K_k .

Three-generation mass formulae.

$$m_k = v \times |L_2|^8 \times \frac{1}{d_1 N^2} \times \lambda_k, \quad k = 1, 2, 3. \quad (13)$$

Particle	L -factor	Node	Gen	Prediction	Experiment	Deviation
ν_3	$ L_2 ^8$	$1/(d_1 N^2)$	λ_3	50.6 meV	~ 50 meV	+1.3%
ν_2	$ L_2 ^8$	$1/(d_1 N^2)$	λ_2	10.0 meV	—	—
ν_1	$ L_2 ^8$	$1/(d_1 N^2)$	λ_1	5.1 meV	—	—

$\Sigma m_\nu = 66$ meV, within CMB-S4 sensitivity (~ 30 meV) [25]. $R = \Delta m_{31}^2 / \Delta m_{21}^2 = 34.5$, consistent with experiment 34.0 ± 1.4 [9] (0.4σ).

CP phase. The CP-violating phase of neutrinos is also determined by the arithmetic properties of χ_3 . The Gauss sum $\tau(\chi_3) = i\sqrt{7}$ is purely imaginary; its argument locks $\delta_{CP} = -\arg(\tau(\chi_3)) = -\pi/2 = -90^\circ$ [A*], consistent with the NOvA/T2K best fit [7, 8].

6 Discussion and Predictions

Sections 2–5 have shown that the 12 fermion masses are completely determined by the character structure of the BC algebra, the Jacobi phase, and the values of Dirichlet L -functions on the critical line, with deviations $\leq 2\%$.

6.1 Summary table

The following consolidates all fermion mass predictions. Each mass is given by the unified formula (§5.3): $m_f = v \times |L|^{\text{pow}} \times \text{node} \times \text{Gen}$.

Particle	Path	L -factor	Node	Gen	Prediction	Experiment	Deviation
τ	$\chi_0 \rightarrow \chi_2$	$ L_2 ^4$	$1/\sqrt{d_1}$	K_0	1791 MeV	1777	+0.8%
μ	$\chi_0 \rightarrow \chi_2$	$ L_2 ^4$	$1/\sqrt{d_1}$	K_1	106.6	105.7	+0.9%
e	$\chi_0 \rightarrow \chi_2$	$ L_2 ^4$	$1/\sqrt{d_1}$	K_2	0.512	0.511	+0.1%
b	$\chi_0 \rightarrow \chi_2 \rightarrow \chi_3$	$ L_2 ^4 L_3$	$d_2/(\sqrt{d_1} \zeta)$	K_0	4219	4180	+0.9%
s	$\chi_0 \rightarrow \chi_2 \rightarrow \chi_3$	$ L_2 ^4 L_3^2$	$1/ \zeta ^2$	K_1	92.9	93.4	−0.5%
d	$\chi_0 \rightarrow \chi_2 \rightarrow \chi_3$	$ L_2 ^4 L_3^2$	$1/ \zeta ^2$	λ^2	4.58	4.67	−2.0%
c	$\chi_0 \rightarrow \chi_2 \rightarrow \chi_3 \rightarrow \chi_1$	$ L_2 ^4 L_3^2 L_1 ^4$	$1/\sqrt{d_1}$	K_0	1273	1270	+0.2%
t	$\chi_0 \rightarrow \chi_2 \rightarrow \chi_3 \rightarrow \chi_1$	$ L_2 ^4 L_3^2 L_1 ^4$	$1/\sqrt{d_1}$	$1/\alpha_{\text{EM}}$	174453	172760	+1.0%
u	$\chi_0 \rightarrow \chi_2 \rightarrow \chi_3 \rightarrow \chi_1$	$ L_2 ^4 L_3^2 L_1 ^4$	$1/\sqrt{d_1}$	$\lambda^2 \alpha_W$	2.1	2.16	−1.9%
ν_3	$\chi_0 \rightarrow \chi_2 \rightarrow \chi_0 \rightarrow \chi_2$	$ L_2 ^8$	$1/(d_1 N^2)$	λ_3	50.6 meV	~50	+1.3%
ν_2		$ L_2 ^8$	$1/(d_1 N^2)$	λ_2	10.0 meV	—	—
ν_1		$ L_2 ^8$	$1/(d_1 N^2)$	λ_1	5.1 meV	—	—

Deviations for all 9 charged fermions are $\leq 2\%$; for neutrinos $\Sigma m_\nu = 66 \text{ meV}$ and $R = 34.5$. All results are derived from 4 L -function values plus $N = \varphi(p^4)$; v , α_{EM} , and α_W are experimental inputs.

6.2 Testable predictions

(1) Neutrinos. This paper makes independent predictions for the absolute masses of all three neutrino generations:

$$m_1 = 5.1 \text{ meV}, \quad m_2 = 10.0 \text{ meV}, \quad m_3 = 50.6 \text{ meV}.$$

m_1 and m_2 have not yet been directly measured; their detection would constitute a direct test. Additional neutrino predictions:

- $\Sigma m_\nu = 66 \text{ meV}$: directly testable by CMB-S4 [25] with sensitivity $\sim 30 \text{ meV}$;
- $R = \Delta m_{31}^2 / \Delta m_{21}^2 = 34.5$: independent of the phase δ and the absolute scale, this is the most robust numerical prediction of the paper. JUNO [13] will measure it to $< 1\%$ precision; the current value $R_{\text{exp}} = 34.0 \pm 1.4$ [9] is consistent at 0.4σ ;

- $\delta_{CP} = -90^\circ$: locked by the Gauss sum $\tau(\chi_3) = i\sqrt{7}$, not a fit. DUNE and Hyper-Kamiokande [14] will measure it to $\pm 5^\circ$;
- Normal ordering (NH): $m_3 > m_2 > m_1$. If JUNO confirms inverted ordering, the framework fails;
- Majorana nature: $\chi_3 = \bar{\chi}_3$ (self-conjugate) requires neutrinos to be Majorana particles. Neutrinoless double-beta decay experiments (nEXO/LEGEND) can test this.

(2) Precise equalities between mass ratios and coupling constants. u-type quarks of all three generations share the same L -factors and node factor; their mass ratios are entirely determined by the inter-generation factors. This yields two precise equalities:

$$\frac{m_t}{m_c} = \frac{1}{\alpha_{\text{EM}}(0)} = 137.036, \quad (14)$$

$$\frac{m_u}{m_c} = \lambda^2 \alpha_W. \quad (15)$$

Equation (14) ties the top/charm mass ratio to the electromagnetic coupling constant: the current value is $m_t/m_c = 172760/1270 = 136.0$, a deviation of 0.8%. The HL-LHC and future e^+e^- colliders will push the precision on top and charm masses to the 0.1% level; if m_t/m_c converges to 137.036 that constitutes strong confirmation, while a departure would falsify the prediction. Equation (15) similarly ties the u /charm ratio to the Cabibbo angle and the weak coupling.

(3) CKM mixing parameters. The three independent CKM elements are uniquely determined by δ_2 and S_3 representation theory:

$$|V_{us}| = 0.222 \text{ (1.1\%)}, \quad |V_{cb}| = 0.040 \text{ (2.3\%)}, \quad |V_{ub}| = 0.00365 \text{ (1.1\%)}. \quad (16)$$

LHCb and Belle II are pushing precision to the 0.1% level.

(4) Higgs mass. Each of the three disjoint doublets of $(\mathbb{Z}/7\mathbb{Z})^*$ is divided by an involution into a symmetric (scalar) branch and an antisymmetric (fermion) branch, each with weight 1/2. The three doublets are independent, and the scalar-branch probabilities multiply:

$$\lambda = \left(\frac{1}{2}\right)^3 = \frac{1}{8}, \quad m_H = v\sqrt{2\lambda} = \frac{v}{2} = 123 \text{ GeV}.$$

The experimental value is 125.1 GeV, a deviation of 1.6%.

(5) Structural predictions.

- **Exactly three generations:** arising from the rigid structure of $\mathbb{Z}/3\mathbb{Z}$. If a fourth generation of fundamental fermions is discovered, the framework fails.

6.3 Open problems

Although this paper has connected the main storylines, several key open problems must be stated honestly:

- **RH/critical-line problem:** This paper repeatedly uses phase-locking on the critical line and the rigidity of finite characters, which suggests that a Riemann-type spectral structure may participate at a deeper level, but the complete connection remains to be established.
- **Origin of $p = 7$:** Why does nature select precisely 7 rather than some other prime? This paper treats 7 as the minimal viable input, but its ultimate origin may require a higher-level selection principle or an absolute-geometry explanation.

6.4 Conclusion

The 12 Standard Model fermion masses—spanning 12 orders of magnitude from the top quark at 173 GeV to neutrinos at ~ 0.05 eV—were previously regarded as independent experimental inputs. This paper presents a unified formula:

$$m_f = v \times |L|^{\text{pow}} \times \text{node factor} \times \text{Gen}, \quad (16)$$

reducing the 12 masses to the values of 4 Dirichlet L -functions on the critical line $s = 1/2$. Deviations for all 9 charged fermions are $\leq 2\%$; the three absolute neutrino masses ($m_3 = 50.6$ meV, $\Sigma m_\nu = 66$ meV) and the mass ratio $R = 34.5$ are derived within the same framework. The precise equalities $m_t/m_c = 1/\alpha_{\text{EM}}$ and $m_u/m_c = \lambda^2 \alpha_W$ further tie the mass spectrum to the force constants.

The $< 1\%$ measurement of R by JUNO (2026–2027), the ~ 30 meV sensitivity on Σm_ν from CMB-S4, and the precise measurements of m_t/m_c and the Higgs self-coupling λ by the HL-LHC, will constitute direct tests of the present framework.

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